Core Model Proposal #390: Base Year Update: Initial updates preparing for the new base year

Product: Global Change Analysis Model (GCAM)

Institution: Joint Global Change Research Institute (JGCRI)

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Date committed: 13 Feb 2025

IR document number:

Related sector: model-wide

Type of development: data, code

Purpose: GCAM's current base year – the last historical year used for calibration – 2015, is considerably lagged compared to the present year, indicating out-of-date assumptions about near-past state of the resources, commodities, and technologies represented in GCAM. This proposal makes data and code changes to enable updating the GCAM base year easier by making the processing code more flexible and robust.

Philosophy

This CMP was motivated by the following goals / broader philosophy for the base year update:

- Write code more defensively, with more checks and, where warranted, automatic correction of problems (with reporting) and try to reduce the number of cryptic failures
- Write code more flexibly, without hard-coding years or assuming a specific relationship between base-year and last year in data (where possible/reasonable), and be consistent with the use of years constants
- Update raw input data wherever necessary for future base year updates e.g., GDP and population, electric generation technology/ATB costs, resource/fossil fuel prices, floor space, US state energy consumption, steel and aluminum sector production and consumption, among others
- Allow use of historical driver data time series (population, GDP, technology costs) after the model base-year, with projections diverging from historical data after the last historical year.
- Document the process so that future base-year updates are relatively less time-consuming and costly.
- Report any data or code update needs for future base year updates or general recommendations to improve gcamdata's robustness

Description of Changes

Summary of Changes

This proposal presents changes in the codebase, updates to raw input data, and sometimes a revision of the processing approach. A detailed item-by-item summary of changes could be found in the **Appendix # 2** and accompanying spreadsheet. The following is a high-level, inexhaustive summary of changes presented in this CMP. We have:

1. Added 5 new countries
2. Extended GDP deflator up to 2022
3. Decommission gSSPs socioeconomics and industry income elasticities
4. Updated ATB technology costs to 2022 (including a new algorithm for keeping temporal consistency)
5. Adopted fossil resource prices from BP and updated data (including breakout into fossil, renewable, & Uranium prices)
6. Updated unconventional oil production in Canada and Venezuela
7. Updated US energy consumption data (SEDS) for GCAM-USA
8. Updated residential and commercial floor space (AEO) for GCAM-USA
9. Updated iron and steel, aluminum and alumina industry data (including a data-relevant revamp)
10. Improved IPCC emissions factors for fugitive CO_2 emissions from fossil fuels

11. Added road emissions estimates and fuel emissions data from GAINS
12. Automated updating of data consistent with changing base years
13. Harmonized the data system to use one constant to control processing related to final calibration years
14. Added a new feature to synthesize chunks and files documentation
15. Improved documentation of multiple chunks
16. Added checks, warnings, error messages throughout

gcamdata R chunk and Code Updates

This section presents major themes of in-code changes. More details could be found in the "Other Updates" section or in Appendix # 2 that presents changes of all files and chunks.

Harmonizing years

We have adopted MODEL_FINAL_BASE_YEAR to control base year related processing throughout the data system. This has impacted roughly more than 50 chunks (details can be seen in Appendix # 2). Currently the data system has two classes of years constants: continuous years and interval years. Continuous years have a yearly time series throughout the model time horizon (HISTORICAL_YEARS starting from 1971 and FUTURE_YEARS starting from HISTORICAL_YEARS+1 through 2100). Interval years have a time-step that was previously standardized to be a 5-year increment. These years include MODEL_BASE_YEARS with a predefined array, MODEL_FINAL_BASE_YEAR as maximum of MODEL_BASE_YEARS, and MODEL_FUTURE_YEARS as a 5-year increment starting from a predefined starting year. In principle continuous years and interval years do not necessarily have to coincide, particularly in time windows closer to the base year.

The upper tail of HISTORICAL_YEARS now represents historical data that could go past MODEL_FINAL_BASE_YEAR, and are no longer required to align perfectly with MODEL_BASE_YEARS. In the previous core-model setup max(HISTORICAL_YEARS) was the same as MODEL_FINAL_BASE_YEAR, however we have now decoupled these to allow using historical data past the base year to improve model behavior in the near-future years. We have changed max(MODEL_BASE_YEARS) to MODEL_FINAL_BASE_YEAR throughout the data system to provide more autonomy over controlling base years through dynamically configurable constants in constants.R.

Other changes include removing hard-code 5-year time steps, such as in zsocio_L2323.iron_steel_Inc_Elas_scenarios.R. and making final calibration year flexible to user input in model_time.cpp. Moreover, years constant are flexible and robust to changing base years, i.e., just changing HISTORICAL_YEARS would automatically change the base year, future years, AgLU years etc., so time shift conditions were removed as they could be controlled using dynamically configurable years constants in constants.R.

Automatic Updating

Several files and code instances (more than 35) have been either updated or automated in the code to interpolate or extrapolate data for the final base year. Previously it was common to extend data within .csv files instead of code, which was fragile and leads to cryptic errors elsewhere in the code. Sometimes the assumptions are indifferent to changing historical years, i.e., either the values are constant or change linearly throughout the model time horizon. Such instances have been automated to be updated with changing base years. For example, technology change percentages for fossil fuels under SSPs (A10.EnvironCost_SSPs used in zenergy_L210.resources.R) is a linear interpolation from zero to the value in 2100, so a code block was introduced to read in just 2100 value and linearly interpolate in the code. This way updating this file remains flexible irrespective of the base year.

Another case for the extrapolation is when grouped data changes over time. This complicates automatic updating and because not all combinations of data items remain the same for all categories over the years. One such example is Land_type_area_ha.csv which does not have "complete" nesting for countries, GLUs and land types, i.e., not all countries have all GLUs and land types in all years. This creates NAs in the final data frame due to irregular combinatorial grouping, even without extrapolation to the latest historical year recent than maximum year in the data, resulting in downstream errors at many places (e.g., solar capacity factors). Ideally, such cases need to be fixed in the data file itself, however, in the meantime, the latest available data year has been copied forward to the final base year, accompanied by a warning written out to inform the user of this assumed extrapolation.

Lastly, all input files that have assumptions for both history and the future inherently assume that the future starts from 2020. Automatic updating, if not done right, could be problematic in specific cases when the base year advances to a year more recent than 2020. These cases include when values change over both history and the future and a separation needs to be maintained e.g., in zwater_L1233.Elec_water.R. A careful deliberation is warranted in such cases to ensure that automatic extrapolation maintains the separation between historical and future years until the assumptions are valid and coincide with the real-world technological state. That's also why every extrapolation where values change over time is accompanied by a warning message or a hard stop depending on the severity of implications of automatic updating.

The table below synthesizes chunks that have the capability to automatically update the files with years in response to changing base years.

Table 1. Summary of chunks that automatically update the final base year.

Chunk	Change
zaglu_L100.0_LDS_preprocessing.R	extended land data system files to the final base year and write a warning

zaglu_L120.LC_GIS_R_LTgis_Yh_GLU.R	extend historical managed forest area through final base year and write a warning
zemissions_L112.ceds_ghg_en_R_S_T_Y.R	added road emissions estimates from GAINS for all historical years; updated GAINS fuel emissions data to incorporate emission factors for the base year; and estimate emissions from alumina for historical years only
zemissions_L231.proc_sector.R	extended prices of unlimited resources to the final base year and write a warning
zenergy_L100.IEA_downscale_ctry.R	filled population shares for some countries to downscale IEA data for all historical years and write a warning
zenergy_L119.solar.R	extended harvested area land types to final base year and write warnings for out-of-data land data and resulting missing data in irradiance potential that affects solar capacity factors
zenergy_L121.liquids.R	copied forward the gas-to-unconventional oil coefficient to the base year and wrote a warning
zenergy_L1231.elec_tech.R	explicitly specified extrapolation method for gas technology efficiency, max improvement and rate of improvement
zenergy_L210.resources.R	automatically extend technology change percentages for fossil fuels under SSPs; automatically interpolate environmental costs under SSPs; flexibly update environmental costs for the final base year under SSP4; interpolate non-energy technology costs and coefficient for fossils for final base year
zenergy_L222.en_transformation.R	automatically interpolate future costs of primary energy transformation technologies; automatically update shareweights of transformation techs and make sure that cellulosic ethanol has zero share-weight in the base year to avoid DDGS calibration errors
zenergy_L270.limits.R	extended renewable resource prices (bio_externality_cost) to final base year
zgcamusa_L143.HDDCDD.R	extrapolate census population data to the final base year

zgcamusa_L144.Commercial.R	extended floorspace from EIA AEO for all historical years including final base year
zgcamusa_L2234.elec_segments.R	extended fuel shares of electricity load segments to all base years
zgcamusa_L2244.nuclear.R	extended nuclear gen II generation by state and year to final base year
zgcamusa_L231.proc_sector.R	extended price information for unlimited resources till final base year
zsocio_L100.GDP_hist.R	extended GDP (MER) from USDA to final base year
zsocio_L180.GDP_macro.R	extended labor force calculations from SSP up to final base year for GCAM Macro
zwater_L110.water_demand_primary.R	extended nuclear efficiency coefficients to all historical years
zwater_L1233.Elec_water.R	extended water shares of cooling system elec gen techs to base year while keeping the separation between historical and future years
zwater_L171.desalination.R	extended EFW IO coefficients to all historical years for desalination
zwater_L173.EFW_manufacturing.R	extended EFW IO coefficients to all historical years for manufacturing water withdrawals
zwater_L174.EFW_municipal.R	extended EFW IO coefficients to all historical years for municipal water withdrawals

Besides years, on few other instances, codebase has been changed to automatically extract constants which were only used at one place in the data system. For example, gcamusa.SEDS_DATA_YEARS constant have been removed and now these years are determined automatically within the zgcamusa_L101.EIA_SEDS.R chunk, accompanied by a warning to notify if SEDS data is outdated compared to final base year.

Defensive Code and Descriptive Debugging Support

One of the goals of this CMP is to make codebase more defensive by introducing checks and error messages to avoid erroneous inputs infiltrating to model outputs. This avoids computational expense and redundancy in debugging, especially if the error could have been stopped/notified at the source.

In most cases where automatic extrapolation is introduced, either a warning or a hard-stop trigger is introduced to inform the user or stop the process depending on the severity of implications resulting from extrapolation, see exampled in Appendix # 1. Warnings have not been introduced in cases where the historical and future data do not change or where a value is needed merely as a placeholder for the base year.

In other cases, error reporting has also been improved by spelling out specific sources of error with values wherever possible. One example is writing increase in base year satiation demand and satiation level to improve error reporting in satiation_demand_function.cpp. Another example is in technology.cpp that now writes absolute difference between model output and calibration value to the main_log.txt to improve calibration error reporting. This helps to identify the direction of the error (negative difference would mean model output is lower and calibration value is higher), which is particularly useful when there are dozens of errors.

Data Updates

Decommissioning gSSP outputs

Up until now, the data system attempted to generate outputs labeled as "SSPs" and "gSSPs". SSP was meant to represent the SSP scenarios as originally deployed, while gSSPs represent SSP assumptions updated with newer values for near-term socioeconomic drivers such as population and near-term GDP projections. Since the recent Core Model Proposal to update the SSP database and a general plan to more regularly update them the need to produce near-term adjustments is no longer needed. Thus, we drop the gSSP outputs. Note that while the Core socioeconomics are just SSP2 we are going to need some way of differentiating it for the Macro total factor productivity which will be different than the true SSP2 scenario. Thus, we create a `socioeconomics_CORE` which is just the same as SSP2 however will pull the CORE TFP assumptions.

As a result, specific outputs related to income elasticities in detailed industry, building and transport sectors have been dropped for standard SSP and GCAM3 scenarios. These income elasticity files are now retained only for gSSPs, and removed from the outputs of all industry, building, and transport sectors.

In total 25 chunks have been impacted by this change, including 6 zsocio_*_Inc_Elas_scenarios.R files and 6 zenergy_xml_*_SSP.R files, among others, see Appendix # 2 for all files and details. Configuration and batch SSP batch configurations (batch_SSP_REF.xml, batch_SSP_SPA1.xml, batch_SSP_SPA23.xml, batch_SSP_SPA4.xml, batch_SSP_SPA5.xml) have also been adapted to read-in the appropriate socioeconomics and industry income elasticities.

Socioeconomics

Five new countries have been added in iso_GCAM_regID.csv to be consistent with the latest changes in population and GDP. gdp_deflator() has been updated up to 2022 in pipeline-helpers.R. Along with updating UN population data, constants socioeconomics.UN_HISTORICAL_YEARS and socioeconomics.FINAL_HIST_YEAR have been removed and instead they are extracted by reading values directly from input files.

Numerous chunks have been modified to incorporate changes to reflect gSSPs. These revisions in socioeconomic outputs have particularly impacted income elasticities within industry, building, and transport sectors, as explained in the decommissioning gSSPs section.

Industry

A data-related revmap of iron and steel and aluminum and alumina detailed industry sectors was required to comply with base year requirements and more transparent processing. For this, new mapping files, new data files and new data sources were added for both sectors. Specifically, the source of aluminum production and consumption data was updated to International Aluminum Institute (IAI) from IAA in IAI_ctry_region.csv. Also, we broke out alumina energy consumption and extended up to 2021 in original data units (TJ) by using Metallurgical Alumina Refining Fuel Consumption category from 1st Jan 1985 to 1st Jan 2022 (all) in alumina_energy_region_IAI.csv. Similarly aluminum energy consumption was also broken out and extended up to 2021 in original data units (GWh) by using Primary Aluminum Smelting Power Consumption category from 1st Jan 1980 to 1st Jan 2022 (all) in aluminum_energy_region_IAI.csv. We also updated country aluminum production up to 2020 from CEDS aluminum_prod_USGS.csv and deleted unused aluminum_prod.csv file for aluminum production. Most of the data pre-processing has been endogenized to avoid undocumented user preprocessing (calculating totals, formatting, unit conversion etc.) in zenergy_L1326.aluminum.R.

For iron and steel, a new mapping file to match World Steel Association country names to GCAM country isos WorldSteelAssociation_country_iso.csv has been added. In the previous core model processing, it was unclear how the shares of iron and steel production from Electric Arc Furnace (EAF) from scrap vs direct rolled iron (DRI) were determined. We have back-calculated these and have added the ratio of Electric Arc Furnace steel production from scrap to total (scrap + DRI) in steel_EAFratio.csv. In addition, Angola's historical steel production (in GCAM region 4) has also been corrected in steel_prod_process.csv.

GCAM-USA

Numerous updates for GCAM-USA have also been made to bring the most recent data in and improve code robustness to changing base years. A few examples include updates to residential and commercial floorspace for all historical years, including final base year, in USA as consistent with EIA AEO 2022 AEO_flsp.csv, zgcamusa_L144.Residential.R, zgcamusa_L144.Commercial.R; updates related to removing hard coded countries and dynamically determining which countries go to "other countries" category in the same aforementioned chunks, and adding filtering in case of partial input data to distinguish between residential and commercial cases in zenergy_L144.building_det_flsp.R. Electricity shares from nuclear for unreasonable values have also been adjusted in zgcamusa_L223.electricity.R.

Another data update is extending state energy consumption data (SEDS) from EIA up to 2020 in EIA_use_all_Bbtu.csv. Other updates include extrapolating census population data to the final base year in zgcamusa_L143.HDDCDD.R, extending fuel shares of electricity load segments to all base years in zgcamusa_L2234.elec_segments.R, extending nuclear gen II generation by state and year to final base year in zgcamusa_L2244.nuclear.R, and extending price information for unlimited resources till final base year in zgcamusa_L231.proc_sector.R. We have also incorporated NEI along with SEDS for determining end year for GCAM-USA SO2 emissions linear control in zgcamusa_L2722.nonghg_elc_linear_control.R. This was required because if either of the data is not updated to a more recent base year, then the end year for linear control does not get written currently in xmls.

Approach Updates

ATB – NREL Annual Technology Baseline

NREL ATB sometimes changes technology names and data between releases, so we adopted a new structure to consistently stitch new data together in historical years.

ATB processing approach has been revamped by breaking out capital, fixed and variable operations and maintenance (O&M) costs into yearly files, by creating workflows to resolve discontinuities and discrepancies within years, and by adding new functions to scale the cost time series based on slope thresholds. All the code changes are contained within the zenergy_L113.atb_cost.R chunk. This update also added separate yearly technology overnight capital cost assumptions, fixed O&M, and variable O&M cost assumptions for the period 2015 through 2050 from NREL 2017 ATB through NREL 2022 ATB – all files are named as NREL_ATB_*.csv.

Other specific changes include deleting ATB capital costs, variable O&M and fixed O&M costs with consolidated years and breaking out into separate files for each year to allow tracking discrepancies in years; updating technology details to be consistent with recent ATB data atb_gcam_mapping.csv; and adding new mapping file to track between ATB technologies in historical ATB datasets ATB_tech_mapping.csv.

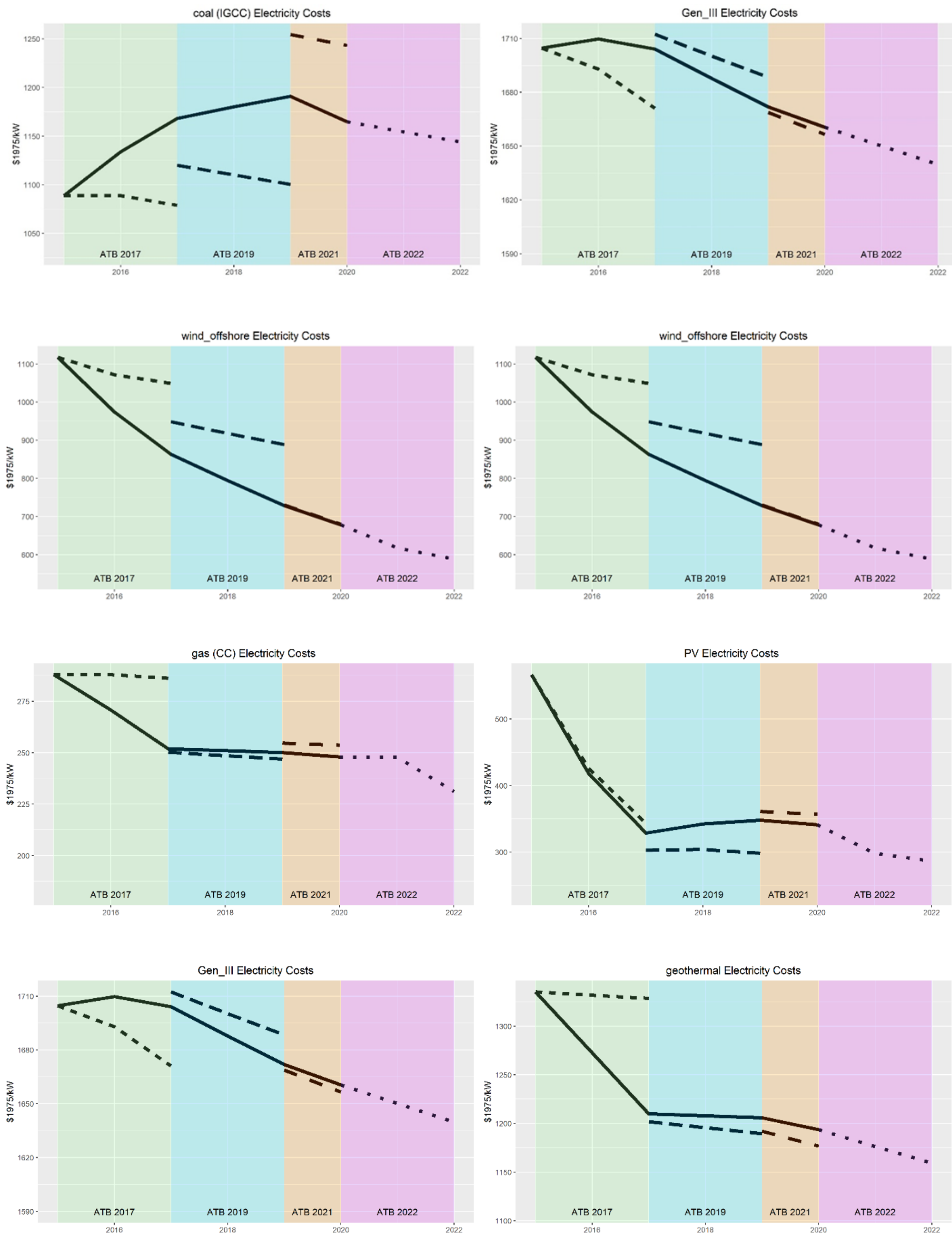


Figure 2. Select examples of new ATB processing approach (full line) to overcome discrepancies of technology cost between ATB data years (dotted lines).

Fuel Resource Prices

Instead of using legacy GCAM 3.1 model inputs, an update to fuel resource prices has been made by shifting to data from the EI Statistical Review of World Energy (formerly BP statistical review), for fossil fuel price timeseries A10.rsrc_info_fossils.csv, formerly contained in A10.rsrc_info.csv together with other resources. Historical Uranium price timeseries A10.rsrc_info_uranium.csv has also been updated based on recent NEA Uranium 2020 Redbook (using US long-term from Fig. 2.10 in 2020 report). Renewable and other resource prices have been broken out into a separate data object A10.rsrc_info_renewables_others.csv but the prices remain largely unchanged.

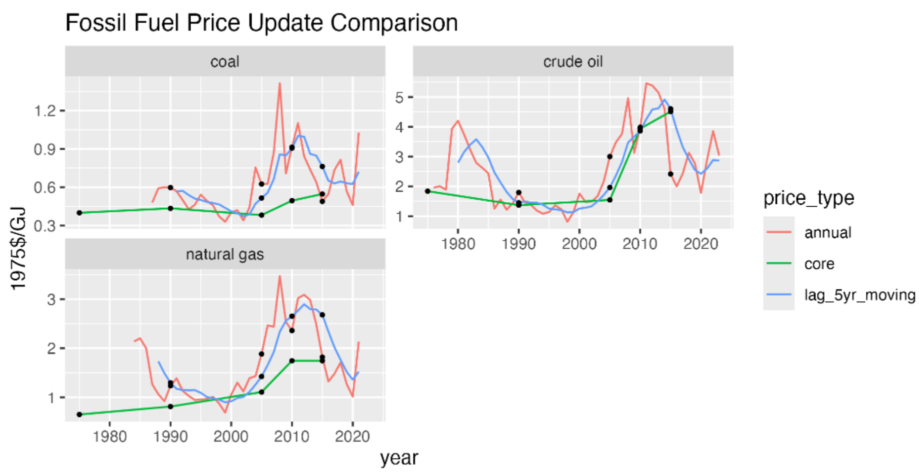


Figure 3. Historical prices of global fuel resource prices (coal, crude oil, and natural gas) in 1975USD/GJ for all resources. The green line represents the previous master, red line represents annual price time series, and the blue line shows updated resource prices in this CMP using a 5-year moving average.

Timeseries for historical fossil fuel prices has been provided for years starting in 1971 through 2021. Coal price timeseries is derived by averaging from two sources (Northwest Europe marker and Japan steam spot CIF), natural gas timeseries uses Average German Import prices, and crude oil uses Brent data. Raw data is kept in original units for comparability and traceability with original data source, and conversion coefficients have been introduced in constants.R and used in zenergy_L210.resources.R to convert from fuel units (barrels, MMBtu etc.) to energy units (EJ, GJ etc.).

Specifically, fuel resource prices update entails:

- splitting rsrc_info file into 3 files for fossils, renewables, and uranium due to varying data sources
- updating coal, oil, gas data source to EI Statistical Review of World Energy (also known as bp-stats-review)
- updating uranium prices according to NEA Red Book 2020

- adding conversion coefficients for oil (barrel to GJ), gas (MMBtu to GJ), and coal (tonne to GJ)
- creating rsrc_info instead of reading in A10.rsrc_info
- reading rsrc_info instead of A10.rsrc_info in all downstream chunks

Some downstream files that were directly reading in the fuel resource prices file (A10.rsrc_info) have also been impacted. For example, zenergy_L1322.Fert.R zenergy_L239.ff_trade.R zgcamusa_L2238.PV_reeds.R zgcamusa_L2239.CSP_reeds.R have been modified inputs to read in endogenously calculated fuel prices data object (L210.rsrc_info).

Other Updates

Energy system

A few other updates in the energy system were also necessary to enable future base year updates. For instance, population shares for some countries to downscale IEA data were filled for all historical years and a warning was introduced in zenergy_L100.IEA_downscale_ctry.R.

For CSP solar energy, a scaling was performed for direct normal irradiance (DNI) and DNI area relative to countries area in zenergy_L119.solar.R. If DNI for areas suitable for CSP is a small fraction of the area of the country, then this will overstate the potential for CSP since most countries will not have access to power generated by CSP. Countries with smaller DNI area to country's total area ratio were scaled to avoid overestimation of average solar capacity for countries such as Russia, Canada etc. for concentrated solar power (CSP) plants.

In the same chunk harvested area land types were extended to final base year and a warning was written for out-of-data land data, resulting in missing data in irradiance potential that affects solar capacity factors.

Data related updates include updating historical unconventional oil production in Canada and Venezuela up to 2021 in rsrc_unconv_oil_prod_bbld.csv. This also comes with an extension of historical gas to unconventional coefficient in the zenergy_L121.liquids.R chunk. Other updates include, updating capital cost of 2nd generation LWR nuclear reactors up to 2020 in A23.globaltech_capital.csv, extending advanced (cheaper) and low (expensive) nuclear capital costs up to 2020 in A23.globaltech_capital_adv.csv and A23.globaltech_capital_low.csv, extending both fixed and variable operations and maintenance costs up to 2020 in A23.globaltech_OMfixed.csv and A23.globaltech_OMvar.csv, and adding electricity technology shareweights for the base year in A23.globaltech_shrwt.csv.

We have improved the logic controlling trade "calibration" between IEA balances and COMTRADE data in zenergy_L2011.ff_ALL_R_C_Y.R. Regardless, a part of this calibration is redundant after CMP # 350 Detailed Natural Gas Trade because there are no NAs or negative trade values in data objects tracking trade flows. Moreover, the code in zenergy_L222.en_transformation.R automatically interpolates future costs of primary energy transformation technologies and automatically updates shareweights of transformation techs.7

Emissions

We have added road emissions estimates from GAINS for all historical years, updated GAINS fuel emissions data to incorporate emission factors for the base year, and estimated emissions from alumina for historical years only in zemissions_L112.ceds_ghg_en_R_S_T_Y.R. Prices for unlimited resources for miscellaneous emissions (A31.rsrc_info.csv) have also been extended to the final base year. This has also been automated in zemissions_L231.proc_sector.R and zgcamusa_L231.proc_sector.R with a warning message. .

Fugitive CO2 emission factors for unconventional oil were also updated in IPCC_unconventional_oil_fug_emfacts.csv and emissions.UNCONVENTIONAL.OIL.FUG.CO2.EMFACT constant was also modified to reflect this change (Bugfix). This improvement to the emission factors was done by excluding "oil sands upgrading" category from the calculation since these would be included in GCAM's existing "refining and processing" emissions not "fugitive" emissions.

Agriculture and Land Use (AgLU)

AgLU years have also been harmonized with dynamically configurable years, so when the base year changes, the relevant AgLU years will also change accordingly. One such example is zaglu_L162.ag_prodchange_R_C_Y_GLU_irr.R where we use AgLU historical years instead of historical years.

We have updated share-weight interpolation in zenergy_L222.en_transformation.R to delay adoption of cellulosic ethanol-based 2G bioenergy, along with an associated code change in zenergy_L222.en_transformation.R to make sure cellulosic ethanol has zero share-weight in the base year to avoid DDGS calibration errors. Other updates include removing negative values for oilcrop and soybean in some regions in recent historical years zaglu_L108.ag_Feed_R_C_Y.R; extending historical managed forest area through final base year and writing a warning, renaming the input file to be consistent with style guidelines zaglu_L120.LC_GIS_R_LTgis_Yh_GLU.R and include first year in calculating land cover change and rates zaglu_L125.LC_tot.R

Water

Energy-for-water input-output coefficients were extended to all historical years for desalination (A71.globaltech_coef.csv), manufacturing, and municipal water withdrawals in zwater_L171.desalination.R zwater_L173.EFW_manufacturing.R zwater_L174.EFW_municipal.R. Water shares of cooling system electricity generation technologies have also been extended to the base year while keeping the separation between historical and future years A23.CoolingSystemShares_RG3.csv. Nuclear efficiency coefficients were also extended to all historical years in zwater_L110.water_demand_primary.R. Finally, South Sudan's iso ssd mapping to Africa continent was also added in mfg_water_mapping.csv.

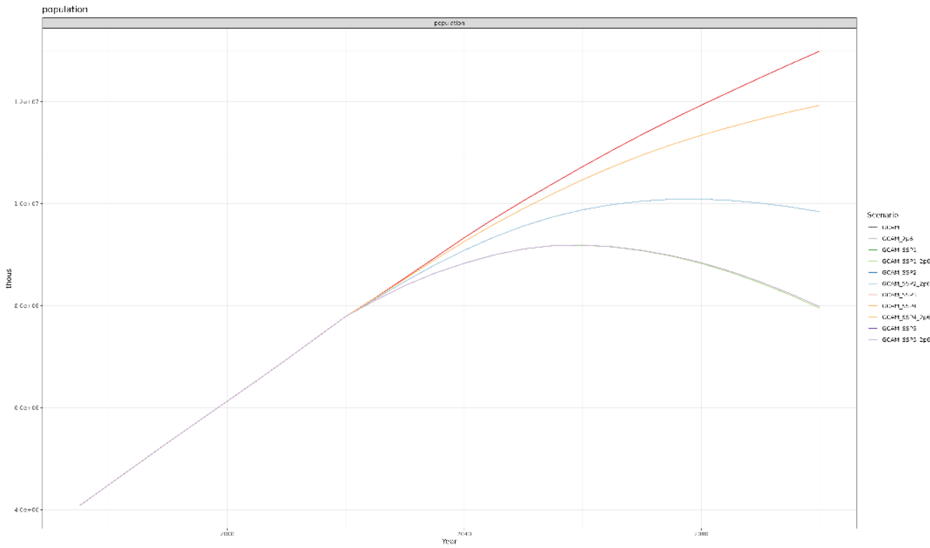
Other minor updates

Other minor updates include adding a "make_drake" make command capability in Makefile to run gcamdata using driver_drake() as a matter of convenience. The previous "xml" target which runs the standard driver() remains unchanged. This is particularly useful while running remote ssh sessions. objects.vcxproj has been modified to change visual studio platform tools version to be consistent with hector's toolset and avoid error in compiling locally on windows. Another issue that has been repeatedly faced by Windows users in running out of file path characters. The solution to debugging MAX_PATH error on windows is as simple as moving the root directory upstream or reducing the characters in the folder name; this has been clarified in driverdrake_vignette.Rmd. Minor updates from constants.R include changed string for disabled modules, add conversion coefficients for fossil fuels, updated fugitive CO2 emission factors for unconventional oil, removed a few constants and endogenized them in relevant chunks e.g., gcamusa.SEDS_DATA_YEARS, gcam.WESTERN_EUROPE_CODE etc.

Validation

All standard validation scenarios (Reference, SSPs, SSP_RCPs, USA, USA_RCP2.6) have solved.

Most changes in this CMP do not change results. However, the changes to the ATB costs primarily, and to a lesser extent primary energy prices, do have some impacts in the energy system.



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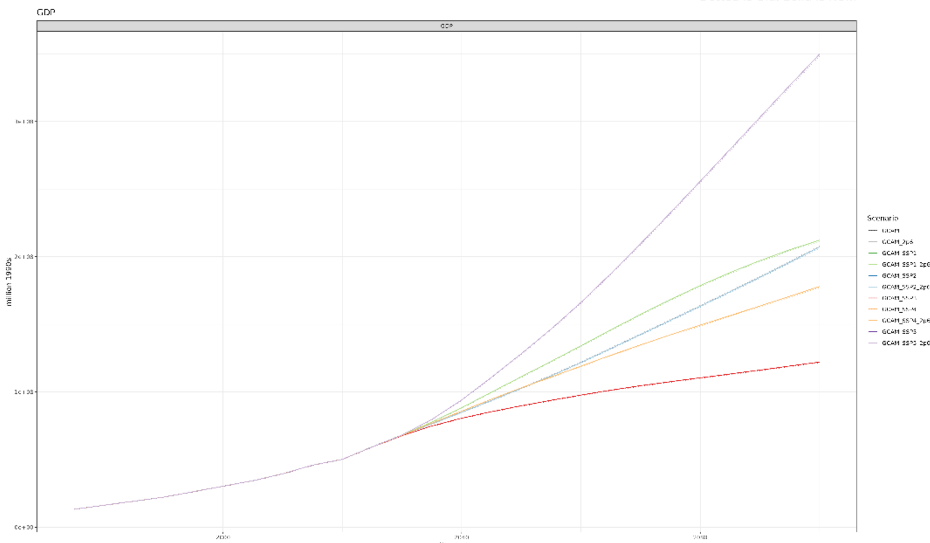

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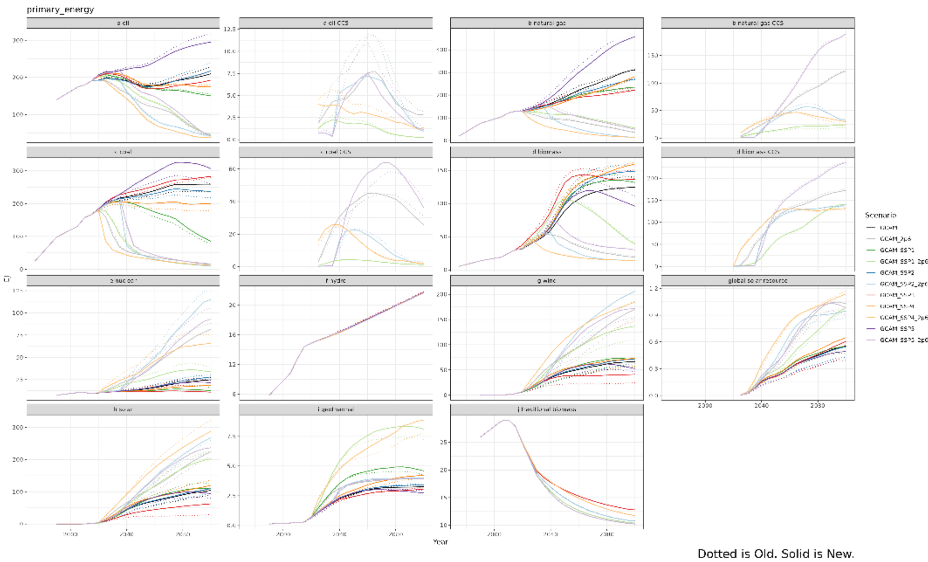
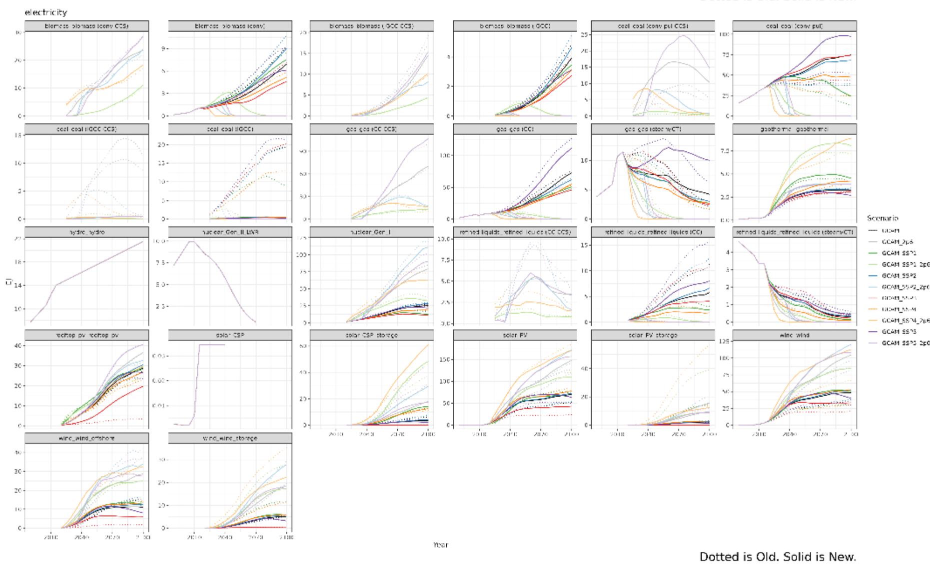
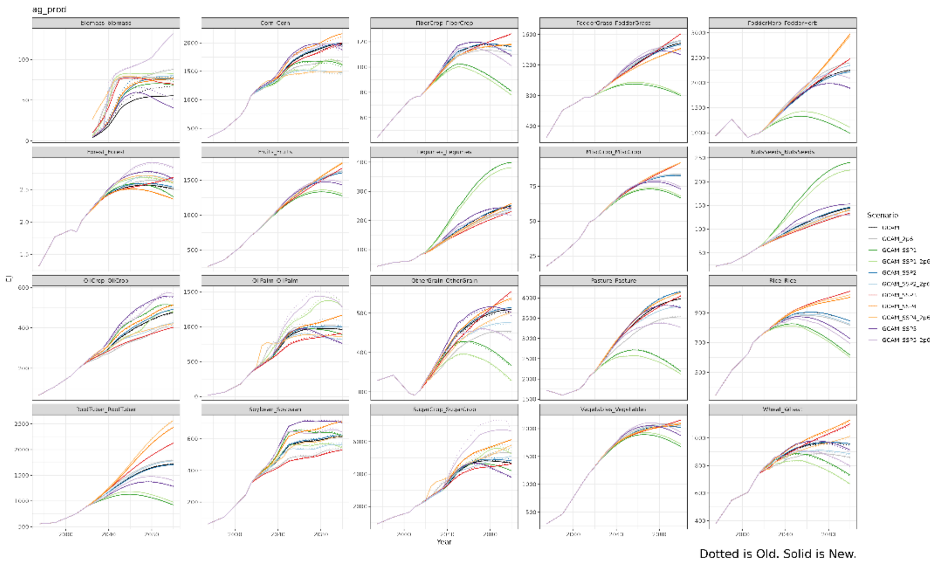
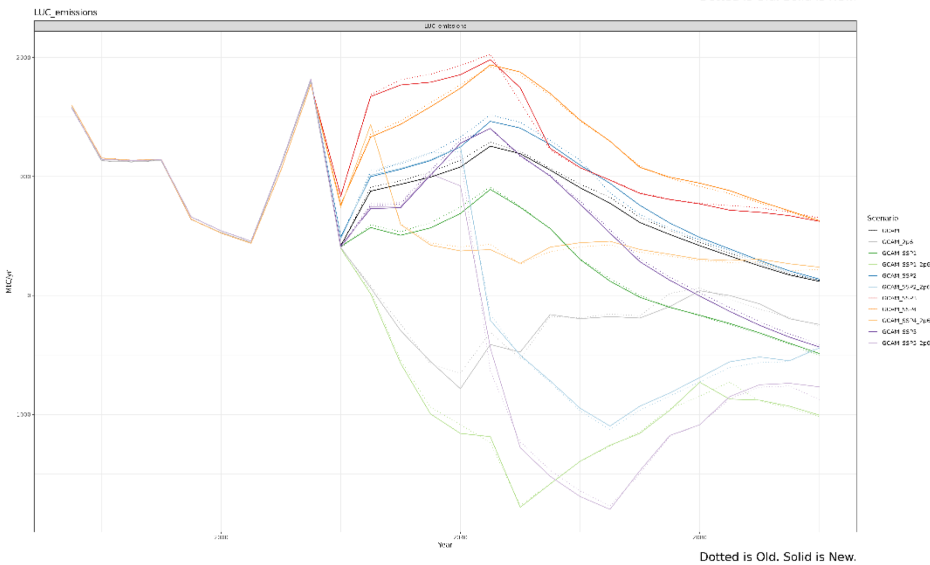


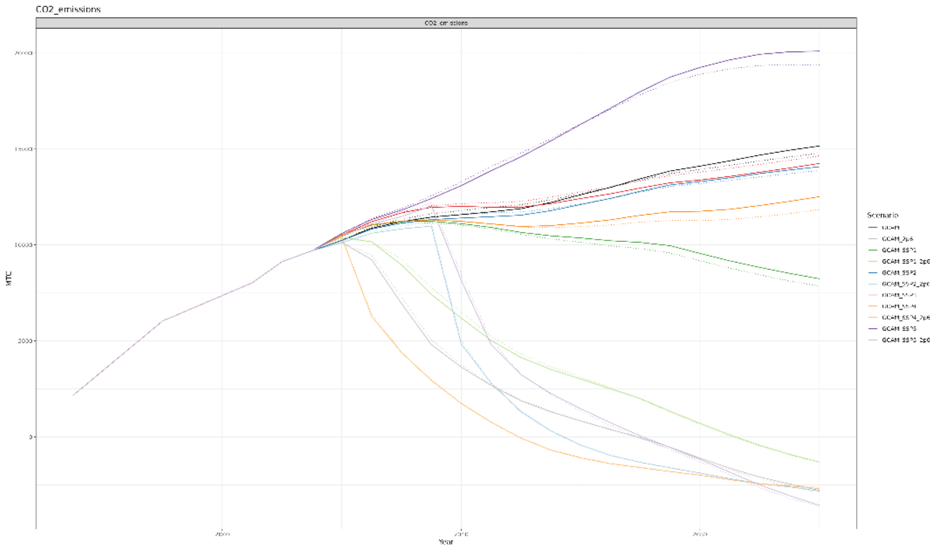


Given no major changes in the Ag and Land use sectors, results there are largely unchanged.

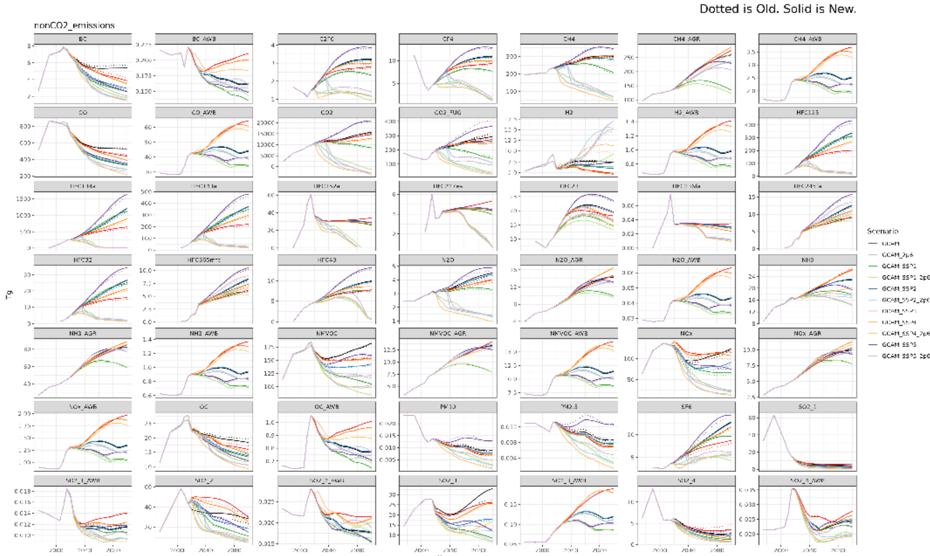




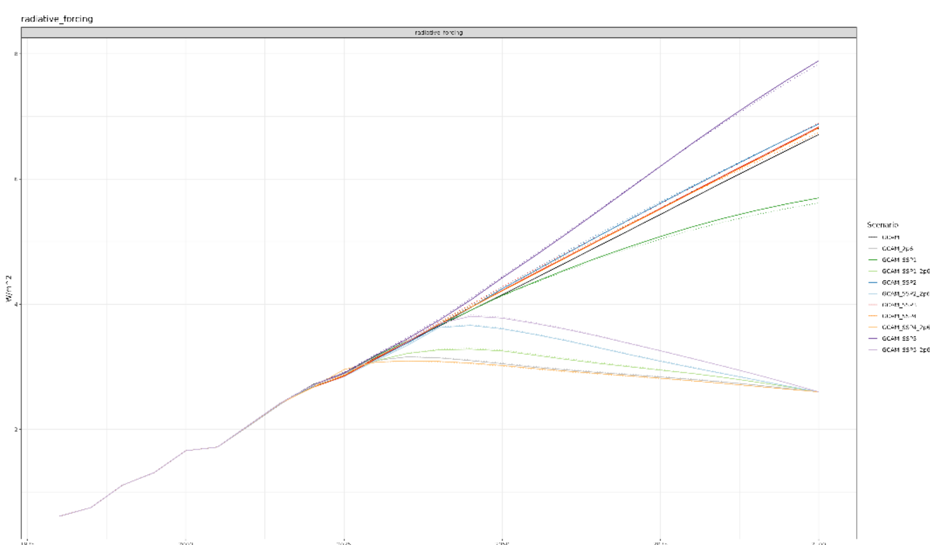
For emissions, while there were some bug fixes related to unconventional oil, most of the differences are a result of energy system changes. Which have higher CO2 emissions in most scenarios. However, also some compensating increase in SO2 emissions leaves the total radiative forcing in a similar place.



Dotted is Old. Solid is New.



Dotted is Old. Solid is New.



Dotted is Old. Solid is New.

Documentation Updates

gcamdata documentation

New documentation feature

A new feature has been added that provides a synthesized overview of description and details of all files, chunks, and outputs of all chunks in the data system.

A new function called chunks_info() has been added in utils.R that generates a synthesized summary file of descriptions and details of all chunks and input files. The chunks_info() function lives in the utils.R script while the overview .csv files. The csv files, CHUNKS_INFO, CHUNKS_INFO_MAP and FILES_INFO, can be generated on demand by using the generate_metadata_csv() function. This functionality will facilitate debugging, onboarding, and improving general ease of working with the data system.

(A)

module	name	description	details	chunk
aglu	module_aglu_L100.0_LDS_preprocessing	Read in and process LDS (Land Data System) files.	Read in the various LDS datasets; regularize their column names and GLU (Geographic Land Unit) data; change Taiwan ISO to that of mainland China; make LDS_ag_HA_ha and LDS_ag_prod_t tables consistent. See Di Vittorio, A., P. Kyle, and W. Collins. 2016. What are the effects of Agro-Ecological Zones and land use region boundaries on land resource projection using the Global Change Assessment Model? Environmental Modelling & Software 85, 246-265. http://dx.doi.org/10.1016/j.envsoft.2016.08.016.	L100.0_LDS_preprocessing
aglu	module_aglu_L100.FAO_preprocessing_OtherData	Get FAO data ready for forestry, fertilizer, animal stock, and land cover	Get FAO data ready for forestry, fertilizer, animal stock, and land cover. Calculate rolling five-year averages.	L100.FAO_preprocessing_OtherData
aglu	module_aglu_L100.FAO_SUA_connection	Pull and further process SUA data needed	Pull and further process SUA data needed. Calculate moving average if needed.	L100.FAO_SUA_connection
aglu	module_aglu_L100.FAO_SUA_PrimaryEquivalent	Gross extraction rate is calculated for domestic, traded, and lagged values. By gross, it means sink items are aggregated. The function is used in Proc_primarize. The function is used in Proc_primarize The function is used in Proc_primarize Scale gross export and import in all regions to make them equal at the world level.	Generate supply utilization balance in primary equivalent This chunk compiles balanced supply utilization data in primary equivalent in GCAM region and commodities. A method to generate primary equivalent is created for the new FAOSTAT supply utilization data (2010 to 2019). New SUA balance is connected to the old one (before 2010). Production and harvested area data with FAO region and item for primary production are provided. For FAO food items, macronutrient values are calculated at SUA item level. Data processing was consistent across scales. Note that GCAM regions and commodities in aggregation mapping can be changed in corresponding mappings. The output data is not averaged over time.	L100.FAO_SUA_PrimaryEquivalent
aglu	module_aglu_L100.GTAP_downscale_ctry	Downscale GTAP region-level land value data to all countries and get GTAP VFA data.	This chunk downscales the GTAP region-level land value to all countries based on production share by GLU and GTAP commodity class.	L100.GTAP_downscale_ctry
aglu	module_aglu_L100.IMAGE_downscale_ctry_yr	Extrapolate IMAGE data to all AGLU historical years and countries.	Each IMAGE table is extrapolated to all AGLU historical years and then downscaled from IMAGE region to all AGLU countries.	L100.IMAGE_downscale_ctry_yr
aglu	module_aglu_L100.regional_ag_an_for_prices	Calculate the calibration prices for all GCAM AGLU commodities.	This chunk calculates average prices over calibration years by GCAM commodity and region. Averages across years, when applicable, are unweighted; averages over FAO item are weighted by production.	L100.regional_ag_an_for_prices
aglu	module_aglu_L101.ag_FAO_R_C_Y	Aggregate FAO food consumption, ag production, and harvested area data to GCAM regions and GCAM commodities.	This chunk aggregates FAO food consumption, agricultural production, and harvested area data up to GCAM commodities and GCAM regions. Data is converted	L101.ag_FAO_R_C_Y

(B)

module	file	units	description	details	filepath
aglu	Land_type_	ha	Land_type_area_ha	area (ha) for land cells in country X glu X land type X protected category X year; Original source: hyde land use areas; reference veg; land cover; country raster; glu raster; hyde land area; Column types: ciiin; ----------; Read from extdata/aglu/LDS/Land_type_area_ha.csv	LDS/Land_type_area_ha
aglu	LDS_ag_HA	ha	Initialization of harvested area (ha) by country/GLU/crop	Initialization of harvested area (ha) by country/GLU/crop; Original source: many, including HYDE and SAGE; Read from extdata/aglu/LDS/LDS_ag_HA_ha.csv	LDS/LDS_ag_HA_ha
aglu	LDS_ag_prc	tons	Initialization of production (t) by country/GLU/crop	Initialization of production (t) by country/GLU/crop; Original source: many, including HYDE and SAGE; Read from extdata/aglu/LDS/LDS_ag_prod_t.csv	LDS/LDS_ag_prod_t
aglu	LDS_value_	million USD	Initialization of land value (million USD) by country87/use/GLU	Initialization of land value (million USD) by country87/use/GLU; Original source: many, including HYDE and SAGE; Read from extdata/aglu/LDS/LDS_value_milUSD.csv	LDS/LDS_value_milUSD
aglu	MIRCA_irrH	ha	MIRCA irrigated harvested area	mirca irrigated harvested area (ha) for sage land cells in country X glu; Original source: MIRCA2000; country raster; new glu raster; Read from extdata/aglu/LDS/MIRCA_irrHA_ha.csv	LDS/MIRCA_irrHA_ha
aglu	MIRCA_rfdI	ha	MIRCA rainfed havested area	mirca rainfed havested area (ha) for sage land cells in country X glu; Original source: MIRCA2000; country raster; new glu raster; Read from extdata/aglu/LDS/MIRCA_rfdHA_ha.csv	LDS/MIRCA_rfdHA_ha
aglu	Mueller_yi	tons/ha	Attainable yield levels by country and basin	Min, average, and max refer to the respective yields observed in grid cells within country, basin, and crop; count is the number of grid cells with harvested area for the given crop, country, and basin.; Original source: Mueller, N. D., Gerber, J. S., Johnston, M., Ray, D. K., Ramankutty, N. and Foley, J. A.: Closing yield gaps through nutrient and water management, Nature, 490(7419), 254?<80><93>257, doi:10.1038/nature11420, 2012. Geospatial data processed by Yannick Le Page.; Read from extdata/aglu/LDS/Mueller_yield_levels.csv	LDS/Mueller_yield_levels
aglu	Ref_veg_ca	Mg/ha	Reference veg and soil C density for HYDE land cells	ref veg soil and veg carbon density (Mg/ha) for hyde land cells in country X glu X land type; Original source: soil c for sage pot veg; veg c for sage pot veg; reference veg; country raster; new glu raster; hyde land area; Read from extdata/aglu/LDS/Ref_veg_carbon_Mg_per_ha.csv	LDS/Ref_veg_carbon_Mg_per_ha
aglu	Water_foot	m3	Average annual water consumption	crop average annual water volume consumed (m^3) for sage land cells in country X glu; Original source: water footprint network; country raster; new glu raster; Read from extdata/aglu/LDS/Water_footprint_m3.csv	LDS/Water_footprint_m3
aglu	AGLU_ctry	NA	Mapping of countries in AGLU databases to ISO codes	Note that both French Guiana and French Guyana are added for now and Saint Helena, Ascension and Tristan da Cunha is added. Area_code should be used later, though they are the same region. Function FAO_AREA_DISAGGREGATE_HIST_DISSOLUTION_ALL deals with region changes over time in gcamdata.; Last update: 4-2-2022 (XZ); Column types: cccciccciccc; ----------; Read from extdata/aglu/AGLU_ctry.csv	AGLU_ctry
common	iso_GCAM_	NA	ISO to GCAM region mapping	Maps iso codes to GCAM regions (including composites as necessary); Column types: ccci; ----------; Read from extdata/common/iso_GCAM_regID.csv	iso_GCAM_regID

Figure 4. Snapshot of the new documentation feature for all chunks in (A) and all data files in (B).

This could help in tracing data flows through the data system without having to run gcamdata.

In-code documentation

Documentation of regenerate documentation of the functions and chunks have been regenerated and new documentation in man/*.Rd has been committed to reflect updates of in-code changes. Spelling corrections in numerous files have been made. In some places, sources of data have also been updated to new links, e.g., updated CDIAC source from ORNL to ESS-DIVE LBL repository in CDIAC_CO2_by_nation.csv.

Recommendations

The updates presented in this CMP should enable updating base year to more recent years with relatively more ease. There are a few other recommendations that could improve the robustness, flexibility, and computational efficiency of gcamdata and GCAM generally.

1. Write absolute and relative difference between model output and calibrated outputs to the debug file. The process of debugging calibration errors becomes tedious and convoluted if an error is observed in a fuel that is demanded by dozens of technologies, that may or may not have a calibrated-value to compare to physical-demand (model output). Debugging is particularly challenging when technology outputs are calibrated by inputs, i.e., calibrated-value = -1, so calibrated-value is then calculated by calOutputValue times the coefficient to compare to physical-demand. This whole process of reconciling calibrated inputs and outputs could be skipped if absolute and relative differences are written to the debug file on technology-input level, providing readily accessible differences between calibrated values and model outputs to debug calibration errors.
2. Remove all GCAM3 processing from gcamdata, starting from L101.Population and L102. GDP. In this CMP we have only removed GCAM3 output xmls and associated processing in chunks producing those xmls. None of the standard configurations use GCAM3, and after this CMP none of the GCAM3 output xmls would be produced anyway so removing GCAM3 processing only offers upside potential for computational efficiency. There are a few mappings that may be required since some assumptions about electricity generating technologies are carried forward from GCAM3 assumptions. Until the raw data is updated to reflect updated assumptions those mappings would be necessary. Other than that, GCAM3 related processing can be stripped off throughout the data system for computational and maintainability gains.
3. Update socioeconomics from new SSPs – new GDP and Population data has been released. This CMP has updated GDP from IMF and population from UN, so these trajectories might differ from standardized SSP scenarios. Many studies, including intermodal comparisons and large ensemble studies, rely on standardized scenarios, so updating to SSP socioeconomics would provide standardized drivers to many model components and benchmarks for comparisons to other models and studies.
4. Abandon standard SSPs altogether and only maintain gSSPs. Standard SSPs has had two releases, first with 2010 as the last historical/base year and second with 2021 as the base year. gSSPs take standard SSPs and modify near-past and near-future GDP and

population assumptions if the base year of the SSPs and GCAM's base year differ. Only maintaining gSSPs would not countermand SSPs, instead it'll give a much more robust and flexible framework to be able keep original SSPs whenever they are released but also keep updating near-past socioeconomic drivers regularly irrespective of standard SSPs releases. Before this CMP the model produced outputs for both SSPs and gSSPs; this CMP removes socioeconomics and industry incomes elasticities for standard SSPs, however removing standard SSPs outputs altogether would greatly reduce computational resources requirements, including both memory (processing speed) and storage (disk space).

5. Decouple HISTORICAL_YEARS and MODEL_FINAL_BASE_YEAR. Many instances in the code use max(HISTORICAL_YEARS) and MODEL_FINAL_BASE_YEAR This limits GCAM's ability to use more recent historical assumptions for things that do not require calibration or things that could improve near-future assumptions. For example, GDP and population do not need to be calibrated.

Appendices

Appendix 1: Excerpts of Recommended Guidelines for Base-year Update Processing

The following are some examples of the guidelines developed in this core model proposal and recommended to the development teams.

Detailed documentation for input files

File: rsrc_unconv_oil_prod_bbld.csv # Title: Historical unconventional oil production by country where applicable # Units: million barrels of oil per day # Source: For Canada use http://earlywarn.blogspot.com/2010/01/tar-sands-production-graph.html until 2010; use StatCan data for years beyond 2010 https://www150.statcan.gc.ca/n1/tbl/csv/25100014-eng.zip for 2016 onwards https://www150.statcan.gc.ca/t1/tbl1/en/tv.action?pid=2510006301; for Venezuela IEA World Energy Outlook 2010 http://www.worldenergyoutlook.org/media/weo2010.pdf; https://iea.blob.core.windows.net/assets/830fe099-5530-48f2-a7c1-11f35d510983/WorldEnergyOutlook2022.pdf # Comments: For Canada: download the data https://www150.statcan.gc.ca/n1/tbl/csv/25100014-eng.zip; filter "Crude Bitumen" and "Synthetic crude oil" both of which qualify as unconventional oil according to the Data Quality Statement: https://www23.statcan.gc.ca/imdb/p2SV.pl?Function=getSurvey&SDDS=2198 or https://www.statcan.gc.ca/eng/statistical-programs/document/2198_D1_T2_V1_B.pdf; for more recent data from 2023 filter "Non-upgraded production of crude bitumen" and "Synthetic crude oil production"; filter "Canada" and sum the VALUES for the year; multiple with VALUE *(6.289/365/1000). Example calculation for 2015; we get roughly 137552 (thousand cubic meters) * 6.28981 (barrels/cubic meter) / 1000 (convert thousands to millions) / 365 (days/year) which gets us 2.37 million barrels of unconventional oil per day. For Venezuela scale unconventional oil production according to total crude oil production (total crude oil production halves from 2015 to 2020)

Put warning and stops wherever appropriate. For example, in zwater_L132.water_demand_manufacturing.R we warn and stop if the mapping files (energy balances countries and manufacturing water withdrawals countries) are mismatched

```
# First check that all needed iso's are in the mapping file and emit an error if not
unique(L101.en_bal_EJ_ctry_Si_Fi_Yh_full$iso) -> df1_iso
unique(mfg_water_mapping$iso) -> df2_iso
setdiff(df1_iso, df2_iso) -> in_df1_not_df2
if (nrow(as.data.frame(in_df1_not_df2)) > 0) {
    stop(paste0("Mapping file mfg_water_mapping does not contain iso's: ",as.character(in_df1_not_df2)))
}
```

Code dynamically wherever possible, for example this code block in zenergy_L210.resources.R automatically reads in the price of year of fossil fuels, dynamically applies the GDP deflator and changes the unit automatically. This is also an example for keeping the raw data as close to the original data source as possible, and endogenizing the data pre-processing in the respective chunks as much as possible.

```
A10.rsrc_info_fossils %>%
      filter(resource == "natural gas") %>%
      na.omit() %>%
      mutate(value = value / CONV_MMBTU_GJ,
             `price-unit` = gsub("millionBtu", "GJ", `price-unit`),
             # the regex parses the currency unit to allow automatic currency deflation to $1975
             value = value * gdp_deflator(1975, as.numeric(unique(unlist(regmatches(`price-unit`,
gregexpr("[[:digit:]]+", `price-unit`)))))),
             `price-unit` = gsub(unique(unlist(regmatches(`price-unit`, gregexpr("[[:digit:]]+", `price-
unit`)))), "1975", `price-unit`)) %>%
      select(!source) -> A10.rsrc_info_fossils_gas
```

Remove recursive assignments of data objects, for example in zenergy_L1323.iron_steel.R

```
92       -     All_steel %>%
93       -       left_join(iso_GCAM_regID, by = "iso") %>%
      91 +     All_steel_Ctry %>%
94    92         #aggregate the production to regional level
95    93         group_by(GCAM_region_ID, year) %>%
96    94         summarise(BLASTFUR=sum(BLASTFUR),`EAF with scrap`=sum(`EAF with scrap`),
97       -                  `EAF with DRI`=sum(`EAF with DRI`))-> All_steel
      95 +                  `EAF with DRI`=sum(`EAF with DRI`))-> All_steel_R
98    96
99    97         #Obtain the index of GCAM_regions and sub sectors that are calibrated to zero steel production in the base-year
100      -       All_steel %>%
      98 +     All_steel_R %>%
101   99           filter(year==MODEL_FINAL_BASE_YEAR & (`EAF with scrap`==0| BLASTFUR==0 | `EAF with DRI`==0)) %>%
102  100           left_join(GCAM_region_names,by=c("GCAM_region_ID"))-> L1323.index
103  101
104  102         #add a minimal steel production value (0.5% of the total) to technologies in the base-year where they are calibrated to zero
105      -       All_steel %>%
     103 +     All_steel_R %>%
```

Automatic updating of data with a warning message if the data changes over time (example from zaglu_L100.0_LDS_preprocessing.R)

```
# base year related processing:
    # check which files have the years column, write a warning if the max year
    # is earlier than the base year, and copy the data to the final base year
    for(nm in namelist) {
      if("year" %in% names(LDSfiles[[nm]])) {
        max_year <- max(LDSfiles[[nm]]$year, na.rm = TRUE)
        # cat(nm, ":", max_year, "\n")

        # Check if the max year is earlier than MODEL_FINAL_BASE_YEAR
        if (max_year < MODEL_FINAL_BASE_YEAR) {
          warning("module_aglu_L100.0_LDS_preprocessing: latest year in ", nm, " (", max_year, ") is earlier
than the final base-year (", MODEL_FINAL_BASE_YEAR, "). Consider updating the data.")
        }

        # Copy latest year data to MODEL_FINAL_BASE_YEAR
        latest_data <- LDSfiles[[nm]] %>%
          filter(year == max_year) %>%
          mutate(year = MODEL_FINAL_BASE_YEAR)
        LDSfiles[[nm]] <- bind_rows(latest_data, LDSfiles[[nm]])
      }
    }
```

Do not assume a 5-year time step, historically or for future, instead determine next year dynamically using arrays of years constants. Below is an example from zsocio_L2323.iron_steel_Inc_Elas_scenarios.R where hard coded 5-year increment was replaced with a function that determines the next year in MODEL_FUTURE_YEARS array.

```
 #Rebuild a new tibble save the previous year value
   L2323.pcgdp_thous90USD_Scen_R_Y_5_before <- L2323.pcgdp_thous90USD_Scen_R_Y %>%
-      mutate(year= year+5,pcgdp_90thousUSD_before = pcgdp_90thousUSD ,steel_cons_before = steel_cons) %>%
-      select(scenario,GCAM_region_ID,region, year,pcgdp_90thousUSD_before,steel_cons_before )
+      # essentially year + 5, but gets the next year dynamically without assuming + 5 year increments
+      mutate(year = sapply(year, function(y) MODEL_FUTURE_YEARS[which(MODEL_FUTURE_YEARS > y)[1]]),
```

Put in-code checks and write a warning or a hard stop depending on the severity of the implications as a result of failed checks. This example from zenergy_L2011.ff_ALL_R_C_Y.R stops gcamdata if global fossil fuel production and consumption are not equal in base years, avoiding downstream miscalculations and erroneous inputs to the model.

```
# check: production should be equal to consumption globally in all historical years
global_cons_Yh <- sum((ff_consumption %>% filter(year %in% MODEL_BASE_YEARS))$consumption)
global_prod_Yh <- sum((ff_production %>% filter(year %in% MODEL_BASE_YEARS))$production)
if (global_cons_Yh != global_prod_Yh) {
    stop("ERROR: Production (", global_prod_Yh, ") and consumption (", global_cons_Yh, ") of fossil fuels in historical years is not equal. ")
}
```

Diagnostic checks could also be left in the code within a FALSE conditional, for example in we check in zwater_L1233.Elec_water.R if the water shares of cooling system electricity generation technologies change over years. This example also demonstrates the need for in-code comments if any years are hard coded.

```
if (FALSE) { # skip running this, but keep here for diagnostics
    shares_combinations_check <- A23.CoolingSystemShares_RG3 %>%
        filter(year <= 2015) %>% # fine to hard code 2015 because we are checking 'historical' years assumed in the file
        group_by(region_GCAM3, plant_type, cooling_system, water_type) %>%
        summarise(all_values_equal = all(value == first(value)))

    # Check if all combinations have the same values for years less than 2015
    if (all(shares_combinations_check$all_values_equal) != T) {
        warning("module_water_L1233.Elec_water: Not all combinations in A23.CoolingSystemShares_RG3 have the same shares for years earlier than 2015. \n")
        # Check using: shares_combinations_check %>% filter(all_values_equal == F)
    }
}
```

Appendix 2: Detailed Summary of Changes in this CMP

Object	Change
cvs/objects/build/vc10	
objects.vcxproj	change visual studio platform tools version to be consistent with hector's toolset and avoid error in compiling locally on windows
cvs/objects/functions/source	
input_accounting.cpp	changed auto_ptr to unique_ptr

satiation_demand_function.cpp	write increase in base year satiation demand and satiation level to improve error reporting
cvs/objects/technologies/source	
technology.cpp	write absolute difference between model and calibrated output to main log to improve calibration error reporting
cvs/objects/util/base/source	
model_time.cpp	made final calibration year flexible to user input
exe	
batch_SSP_REF.xml	adapt SSP batch configurations to read-in gSSP socioeconomics and industry income elasticities
batch_SSP_SPA1.xml	adapt SSP batch configurations to read-in gSSP socioeconomics and industry income elasticities
batch_SSP_SPA23.xml	adapt SSP batch configurations to read-in gSSP socioeconomics and industry income elasticities
batch_SSP_SPA4.xml	adapt SSP batch configurations to read-in gSSP socioeconomics and industry income elasticities
batch_SSP_SPA5.xml	adapt SSP batch configurations to read-in gSSP socioeconomics and industry income elasticities
input/gcamdata/data	
CHUNKS_INFO.csv	introduced a new file that contains a summary of descriptions and details of all chunks
CHUNKS_INFO_MAP.csv	introduced a new file that contains a summary of descriptions and details of all chunks, along with the documentation of its outputs
FILES_INFO.csv	introduced a new file that contains a summary of descriptions and details of all input files in the data system

GCAM_DATA_MAP.rda	binary file, reflects updates in data flows within modules
PREBUILT_DATA.rda	binary file, reflects updates to proprietary data
input/gcamdata/data-raw	
generate_package_data.R	new code that generates CHUNKS_INFO, CHUNKS_INFO_MAP and FILES_INFO along with package data whenever there is a change in input files, data flows, or chunk documentation
inst/extdata/input/gcamdata/aglu/LDS	
LC_bm2_R_MgdFor_Yh_GLU_beforeadjust.csv	renamed the file name to be consistent with the naming convention
inst/extdata/input/gcamdata/common	
iso_GCAM_regID.csv	added 5 new countries to be consistent with latest changes in population and GDP
inst/extdata/input/gcamdata/emissions	
A31.rsrc_info.csv	extended unlimited prices for miscellaneous emissions to the latest base year. This has also been automated in module_emissions_L231.proc_sector and module_gcamusa_L231.proc_sector with a warning message
CDIAC_CO2_by_nation.csv	updated CDIAC source from ORNL to ESS-DIVE LBL repository
IPCC_unconventional_oil_fug_emfacts.csv	updated fugitive CO2 emission factors for unconventional oil
inst/extdata/input/gcamdata/energy/mappings	
atb_gcam_mapping.csv	update technology details to be consistent with recent ATB data
ATB_tech_mapping.csv	added new mapping file to track between ATB technologies in historical ATB datasets
IAI_ctry_region.csv	update source to International Aluminum Institute (IAI) from IAA

WorldSteelAssociation_country_iso.csv	added new mapping file to match World Steel Association country names to GCAM country isos
inst/extdata/input/gcamdata/energy	
A10.rsrc_info_fossils.csv	shifted to BP for fossil fuel price time series instead of GCAM 3.1 model inputs
A10.rsrc_info_renewables_others.csv	renewable and other resource prices broken out into a separate data object
A10.rsrc_info_uranium.csv	updated historical uranium price time series based on the recent NEA Uranium 2020 red book
A22.globaltech_shrwt.csv	updated share-weight interpolation to delay adoption of cellulosic ethanol-based 2G bioenergy
A23.globaltech_capital.csv	updated capital cost of 2nd generation LWR nuclear reactors up to 2020
A23.globaltech_capital_adv.csv	extended advanced (cheaper) nuclear capital costs up to 2020
A23.globaltech_capital_low.csv	extended low (expensive) nuclear capital costs up to 2020
A23.globaltech_eff.csv	removed non-significant zeros
A23.globaltech_OMfixed.csv	extended fixed operations and maintenance costs up to 2020
A23.globaltech_OMvar.csv	extended variable operations and maintenance costs up to 2020
A23.globaltech_shrwt.csv	added electricity technology shareweights for the base year
A_irnstl_RegionalSubsector.csv	changed 2015 to final base year in comments
alumina_energy_region_IAI.csv	changed the source, broke out alumina energy consumption and extended up to 2021 in original data units (TJ)
aluminum_energy_region_IAA.csv	deleted the file and broke out alumina and aluminum energy consumption as they have different data sources

aluminum_energy_region_IAI.csv	changed the source, broke out aluminum energy consumption and extended up to 2021 in original data units (GWh)
aluminum_prod.csv	deleted unused file for aluminum production
aluminum_prod_region_IAI.csv	changed the source and renamed the file
aluminum_prod_USGS.csv	updated country aluminum production up to 2020 from CEDS
NREL_ATB_capital.csv	deleted ATB capital cost file with consolidated years and broke out into separate files for each year to allow tracking discrepancies in years
NREL_ATB_capital_2017.csv	added technology overnight capital cost assumptions for 2015-2050 from NREL 2017 ATB
NREL_ATB_capital_2019.csv	added technology overnight capital cost assumptions for 2017-2050 from NREL 2019 ATB
NREL_ATB_capital_2021.csv	added technology overnight capital cost assumptions for 2019-2050 from NREL 2021 ATB
NREL_ATB_capital_2022.csv	added technology overnight capital cost assumptions for 2020-2050 from NREL 2022 ATB
NREL_ATB_OMfixed.csv	deleted ATB fixed operations and maintenance costs file with consolidated years and broke out into separate files for each year to allow tracking discrepancies in years
NREL_ATB_OMfixed_2017.csv	added technology fixed operations and maintenance cost assumptions for 2015-2050 from NREL 2017 ATB
NREL_ATB_OMfixed_2019.csv	added technology fixed operations and maintenance cost assumptions for 2017-2050 from NREL 2019 ATB
NREL_ATB_OMfixed_2021.csv	added technology fixed operations and maintenance cost assumptions for 2019-2050 from NREL 2021 ATB

NREL_ATB_OMfixed_2022.csv	added technology fixed operations and maintenance cost assumptions for 2020-2050 from NREL 2022 ATB
NREL_ATB_OMvar.csv	deleted ATB variable operations and maintenance costs file with consolidated years and broke out into separate files for each year to allow tracking discrepancies in years
NREL_ATB_OMvar_2017.csv	added technology variable operations and maintenance cost assumptions for 2015-2050 from NREL 2017 ATB
NREL_ATB_OMvar_2019.csv	added technology variable operations and maintenance cost assumptions for 2017-2050 from NREL 2019 ATB
NREL_ATB_OMvar_2021.csv	added technology variable operations and maintenance cost assumptions for 2019-2050 from NREL 2021 ATB
NREL_ATB_OMvar_2022.csv	added technology variable operations and maintenance cost assumptions for 2020-2050 from NREL 2022 ATB
rsrc_unconv_oil_prod_bbld.csv	updated historical unconventional oil production in Canada and Venezuela up to 2021
steel_EAFratio.csv	added ratio of Electric Arc Furnace steel production from scrap to total (scrap + DRI)
steel_prod_process.csv	corrected Angola's historical steel production (in GCAM region 4)
inst/extdata/input/gcamdata/gcam-usa	
AEO_2015_flsp.csv	deleted the file with hard-titled base year
AEO_flsp.csv	updated residential and commercial floorspace in USA consistent with AEO 2022
EIA_use_all_Bbtu.csv	updated state energy consumption data (SEDS) from EIA up to 2020
inst/extdata/input/gcamdata/socioeconomics	

IMF_GDP_growth.csv	updated near-term GDP growth rates from IMF up to 2019 to update SSPs
UN_Pop_Total.csv	updated historical population from UN up to 2022 with projections until 2100 to update SSPs
inst/extdata/input/gcamdata/water	
A23.CoolingSystemShares_RG3.csv	extended water shares of cooling system elec gen techs to base year while keeping the separation between historical and future years
A71.globaltech_coef.csv	extended energy-for-water coefficients for desalination
basin_water_demand_1990_2015.csv	Added a note that the file is not used in the data system but is useful for water-related analysis
mfg_water_mapping.csv	added South Sudan ssd mapping to Africa continent
inst/extdata/input/gcamdata/man	regenerate documentation of the functions and chunks
inst/extdata/input/gcamdata/R	
admin.R	spelling change
constants.R	changed string for disabled modules; made year constant flexible and robust to changing base years; add conversion coefficients for fossil fuels; updated fugitive CO2 emission factors for unconventional oil; removed a few constants and endogenized them in relevant chunks e.g., gcamusa.SEDS_DATA_YEARS, gcam.WESTERN_EUROPE_CODE etc.; removed time shift conditions as year constants are dynamically configurable
module-helpers.R	adopted MODEL_FINAL_BASE_YEAR and gSSPs
pipeline-helpers.R	updated gdp_deflator up to 2022
sysdata.rda	binary file, reflects LEVEL2_DATA_NAMES

utils.R	new function that returns description and details of a chunk. It's used in generate_package_data.R to generate CHUNKS_INFO, CHUNKS_INFO_MAP and FILES_INFO
zaglu_L100.0_LDS_preprocessing.R	extended land data system files to the final base year and write a warning
zaglu_L101.ag_FAO_R_C_Y.R	removed unused harvested area land type data object
zaglu_L108.ag_Feed_R_C_Y.R	removed negative values for oilcrop and soybean in some regions in recent historical years
zaglu_L109.ag_an_ALL_R_C_Y.R	specified source package tidyr for gather
zaglu_L112.ag_prodchange_R_C_Y_GLU.R	corrected comments to avoid referring to a certain year as base year
zaglu_L120.LC_GIS_R_LTgis_Yh_GLU.R	extended historical managed forest area through final base year and write a warning, renamed the input file to be consistent with style guidelines
zaglu_L123.LC_R_MgdPastFor_Yh_GLU.R	spelling change
zaglu_L125.LC_tot.R	include first year in calculating land cover change and rates
zaglu_L162.ag_prodchange_R_C_Y_GLU_irr.R	use AgLU historical years instead of historical years
zaglu_L202.an_input.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zaglu_L203.ag_an_demand_input.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zaglu_L2062.ag_Fert_irr_mgmt.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zaglu_L221.land_input_1.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zaglu_L222.land_input_2.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zaglu_L2231.land_input_3_irr.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing

zaglu_L2252.land_input_5_irr_mgmt.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zemissions_L112.ceds_ghg_en_R_S_T_Y.R	added road emissions estimates from GAINS for all historical years; updated GAINS fuel emissions data to incorporate emission factors for the base year; and estimate emissions from alumina for historical years only
zemissions_L161.nonghg_en_ssp_R_S_T_Y.R	adopted gSSPs; dynamically select EU-15 region code and remove the gcam.WESTERN_EUROPE_CODE constant
zemissions_L2112.ag_nonco2_IRR_MGMT.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zemissions_L231.proc_sector.R	extended prices of unlimited resources to the final base year and write a warning
zenergy_L100.IEA_downscale_ctry.R	filled population shares for some countries to downscale IEA data for all historical years and write a warning
zenergy_L1011.fossilFuel_GrossTrade_EJ_R_C_Y.R	silence package check notes
zenergy_L113.atb_cost.R	revamped ATB processing approach by breaking out capital, OMfixed, and OMvar costs into yearly files; created workflows to resolve discontinuities and discrepancies within years; new functions to scale the cost timeseries based on slope thresholds etc.
zenergy_L115.roofPV.R	spelling change
zenergy_L119.solar.R	scale direct normal irradiance (DNI) and DNI area to countries with smaller DNI area to country's total area ratio to avoid overestimation of solar capacity, especially for concentrated solar power (CSP) plants; also extend harvested area land types to final base year and write warnings for out-of-data land data and resulting missing data in irradiance potential that affects solar capacity factors
zenergy_L120.offshore_wind.R	filter offshore wind OM fixed cost flexibly

zenergy_L121.liquids.R	copied forward the gas-to-unconventional oil coefficient to the base year and wrote a warning
zenergy_L122.gasproc_refining.R	removed redundant code, spelling changes
zenergy_L123.electricity.R	spelling change
zenergy_L1231.elec_tech.R	explicitly specified extrapolation method for gas technology efficiency, max improvement and rate of improvement
zenergy_L132.industry.R	spelling change
zenergy_L1321.cement.R	added an error flag
zenergy_L1322.Fert.R	modified inputs due to endogenously calculated fuel prices data object (L210.rsrc_info)
zenergy_L1323.iron_steel.R	remove recursive assignments of data objects
zenergy_L1326.aluminum.R	change aluminum production and consumption data source; endogenize data processing to avoid undocumented user-preprocessing (calculating totals, formatting, unit conversion etc.)
zenergy_L144.building_det_flsp.R	removed hard coded countries and dynamically determine which countries go to "other countries" category; and filters to distinguish between resid and comm cases, in case of partial input data
zenergy_L162.dac.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zenergy_L2011.ff_ALL_R_C_Y.R	corrected trade "calibration" between IEA balances and COMTRADE data
zenergy_L210.resources.R	changed resource prices; endogenized data processing; automatically extend technology change percentages for fossil fuels under SSPs; automatically interpolate environmental costs under SSPs; flexibly update environmental costs for the final base year under SSP4; interpolate non-energy technology costs and coefficient for fossils for final base year

zenergy_L221.en_supply.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zenergy_L221.en_supply.R	automatically interpolate future costs of primary energy transformation technologies; automatically update shareweights of transformation techs and make sure that cellulosic ethanol has zero share-weight in the base year to avoid DDGS calibration errors
zenergy_L223.electricity.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zenergy_L2231.wind_update.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing, and added a precursor
zenergy_L225.hydrogen.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing; removed redundant code
zenergy_L226.en_distribution.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zenergy_L232.other_industry.R	adopted MODEL_FINAL_BASE_YEAR and gSSPs; removed industry income elasticities for GCAM3 and SSP scenarios (keep only gSSPs)
zenergy_L2321.cement.R	removed industry income elasticities for GCAM3 and SSP scenarios (keep only gSSPs)
zenergy_L2322.Fert.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing; removed redundant code
zenergy_L2323.iron_steel.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing; removed redundant code
zenergy_L2324.Off_road.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing; removed redundant code
zenergy_L2325.chemical.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing; removed redundant code

zenergy_L2326.aluminum.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing; removed redundant code
zenergy_L239.ff_trade.R	added resource prices data object
zenergy_L2392.gas_trade.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zenergy_L242.building_agg.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zenergy_L244.building_det.R	adopted MODEL_FINAL_BASE_YEAR and gSSPs
zenergy_L262.dac.R	adopted MODEL_FINAL_BASE_YEAR and gSSPs
zenergy_L270.limits.R	extended renewable resource prices (bio_externality_cost) to final base year
zenergy_xml_aluminum_SSP.R	removed industry income elasticities for GCAM3 and SSP scenarios (keep only gSSPs)
zenergy_xml_cement_SSP.R	removed industry income elasticities for GCAM3 and SSP scenarios (keep only gSSPs)
zenergy_xml_chemical_SSP.R	removed industry income elasticities for GCAM3 and SSP scenarios (keep only gSSPs)
zenergy_xml_iron_steel_SSP.R	removed industry income elasticities for GCAM3 and SSP scenarios (keep only gSSPs)
zenergy_xml_negative_emissions_budget.R	removed GCAM3 and SSP scenarios from comments
zenergy_xml_Off_road_SSP.R	removed industry income elasticities for GCAM3 and SSP scenarios (keep only gSSPs)
zenergy_xml_other_industry_SSP.R	removed industry income elasticities for GCAM3 and SSP scenarios (keep only gSSPs)

zgcamusa_L101.EIA_SEDS.R	removed gcamusa.SEDS_DATA_YEARS constant, determine those automatically; added a warning to notify if SEDS is outdated
zgcamusa_L103.water_mapping.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zgcamusa_L120.offshore_wind_reeds.R	filter offshore wind OM fixed cost flexibly
zgcamusa_L143.HDDCDD.R	extrapolate census population data to the final base year
zgcamusa_L144.Commercial.R	extended commercial floorspace from EIA AEO for all historical years including final base year
zgcamusa_L144.Residential.R	updated residential and commercial floorspace in USA consistent with AEO 2022
zgcamusa_L203.water_td.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zgcamusa_L222.en_transformation.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zgcamusa_L223.electricity.R	adjust electricity shares from nuclear for unreasonable values; adopted MODEL_FINAL_BASE_YEAR
zgcamusa_L2232.electricity_FERC.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zgcamusa_L2232.electricity_FERC.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zgcamusa_L2234.elec_segments.R	extended fuel shares of electricity load segments to all base years
zgcamusa_L2235.elec_segments_FERC.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zgcamusa_L2237.wind_reeds.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zgcamusa_L2238.PV_reeds.R	added resource prices data object; adopted MODEL_FINAL_BASE_YEAR
zgcamusa_L2239.CSP_reeds.R	added resource prices data object; adopted MODEL_FINAL_BASE_YEAR

zgcamusa_L2241.coal_retire.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zgcamusa_L2244.nuclear.R	extended nuclear gen II generation by state and year to final base year
zgcamusa_L2245.geothermal_fixed.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zgcamusa_L231.proc_sector.R	extended price information for unlimited resources till final base year
zgcamusa_L232.industry.R	removed industry income elasticities for GCAM3 in GCAM USA
zgcamusa_L232.water_demand_industry.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zgcamusa_L2321.cement.R	removed industry income elasticities for GCAM3 in GCAM USA
zgcamusa_L2322.Fert.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zgcamusa_L2322.industry_vintage.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zgcamusa_L244.building.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zgcamusa_L270.limits.R	removed industry income elasticities for GCAM3 and SSP scenarios (keep only gSSPs)
zgcamusa_L2722.nonghg_elc_linear_control.R	incorporated NEI along with SEDS for determining end year for GCAM-USA SO2 emissions linear control
zgcamusa_L277.nonghg_prc.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zgcamusa_xml_cement.R	removed industry income elasticities for GCAM3 in GCAM USA
zgcamusa_xml_industry.R	removed industry income elasticities for GCAM3 in GCAM USA
zgcamusa_xml_industry.R	adopted MODEL_FINAL_BASE_YEAR to control base year related processing
zsocio_L100.GDP_hist.R	extended GDP (MER) from USDA to final base year

zsocio_L100.Population_downscale_ctry.R	updated UN population data; removed socioeconomics.UN_HISTORICAL_YEARS and socioeconomics.FINAL_HIST_YEAR and instead extract them by reading values directly from input files; modified the way; updated historical population from UN up to 2022 with projections until 2100 to update SSPspopulation data is made - uses most recent UN historical population data wherever possible, fills in the remaining future years with SSP projections
zsocio_L101.Population.R	updated UN population data; removed socioeconomics.UN_HISTORICAL_YEARS and socioeconomics.FINAL_HIST_YEAR and instead extract them by reading values directly from input files; modified the way; updated historical population from UN up to 2022 with projections until 2100 to update SSPspopulation data is made - uses most recent UN historical population data wherever possible, fills in the remaining future years with SSP projections
zsocio_L102.GDP.R	updated near-term GDP growth rates from IMF up to 2019 and update SSPs processing
zsocio_L180.GDP_macro.R	extended labor force calculations from SSP up to final base year for GCAM Macro
zsocio_L201.Pop_GDP_scenarios.R	adopted gSSPs and data table name change
zsocio_L2323.iron_steel_Inc_Elas_scenarios.R	made processing base year friendly; removed industry income elasticities for GCAM3 and SSP scenarios (keep only gSSPs)
zsocio_L2324.Off_road_Inc_Elas_scenarios.R	removed industry income elasticities for GCAM3 and SSP scenarios (keep only gSSPs)

zsocio_L2325.chemical_Inc_Elas_scenarios.R	removed industry income elasticities for GCAM3 and SSP scenarios (keep only gSSPs)
zsocio_L2326.aluminum_Inc_Elas_scenarios.R	made processing base year friendly; removed industry income elasticities for GCAM3 and SSP scenarios (keep only gSSPs)
zsocio_L242.Bld_Inc_Elas_scenarios.R	removed building sector income elasticities for GCAM3 and SSP scenarios (keep only gSSPs)
zsocio_L252.Trn_Inc_Elas_scenarios.R	removed transport sector income elasticities for GCAM3 and SSP scenarios (keep only gSSPs)
zsocio_xml_bld_agg.R	removed building sector income elasticities for GCAM3 and SSP scenarios (keep only gSSPs)
zsocio_xml_socioeconomics_SSP.R	removed socioeconomic outputs for GCAM3 and SSP scenarios (keep only gSSPs)
zwater_L101.water_supply_groundwater.R	silence package check notes
zwater_L101.water_supply_groundwater.R	extended nuclear efficiency coefficients to all historical years
zwater_L1233.Elec_water.R	extended water shares of cooling system elec gen techs to base year while keeping the separation between historical and future years
zwater_L132.water_demand_manufacturing.R	warn and stop if the mapping file are mismatched
zwater_L171.desalination.R	extended EFW IO coefficients to all historical years for desalination
zwater_L173.EFW_manufacturing.R	extended EFW IO coefficients to all historical years for manufacturing water withdrawals
zwater_L174.EFW_municipal.R	extended EFW IO coefficients to all historical years for municipal water withdrawals
zwater_L201.water_resources_constrained.R	renamed basin water demands to reflect flexible base year

zwater_L202.water_resources_unlimited.R	spelling change
zwater_L203.water_td.R	adopted MODEL_FINAL_BASE_YEAR as a constant to control final calibration year
zwater_L232.water_demand_manufacturing.R	adopted MODEL_FINAL_BASE_YEAR as a constant to control final calibration year
zwater_L270.EFW_input_coefs.R	adopted MODEL_FINAL_BASE_YEAR as a constant to control final calibration year
zwater_L273.EFW_manufacturing.R	adopted MODEL_FINAL_BASE_YEAR as a constant to control final calibration year
zwater_L274.EFW_municipal.R	adopted MODEL_FINAL_BASE_YEAR as a constant to control final calibration year
zwater_xml_electricity_water.R	spelling change
inst/extdata/input/gcamdata/tests/testthat	
test_module_helpers.R	adopted MODEL_FINAL_BASE_YEAR as a constant to control final calibration year
inst/extdata/input/gcamdata/tests/testthat	
driverdrake_vignette.Rmd	clarify the solution to MAX_PATH error on windows
inst/extdata/input/gcamdata	
.gitignore	ignore tracking proprietary data
NAMESPACE	functions and packages list
Makefile	added the capability to run gcamdata using drake with and without generating outputs